2. Findings (Exploratory Data Analysis)

This section answers the 20 EDA questions from the project brief. Each answer is based on the notebook outputs and charts, using the cleaned dataset (250,000 rows) unless stated otherwise.

Question numbers follow the project brief. Where the notebook used a different number, the brief numbering is used here.

2.1 Price and size analysis

Q1. What is the distribution of property prices? Prices range from about ₹15 lakh to ₹495 lakh, with a mean of ₹254.6 lakh and a median of ₹253.9 lakh. The middle 50% of properties lie between ₹132.6 lakh and ₹376.9 lakh. The histogram is almost flat across this range, meaning every price band has roughly the same number of properties. The taller bars at both ends come from capping extreme values at the 1st and 99th percentile during cleaning (see Section 1.2).

Q2. What is the distribution of property sizes? Sizes range from 545 to 4,956 sq ft, with a median of 2,747 sq ft. The middle 50% lies between 1,623 and 3,874 sq ft. As with price, the histogram is nearly flat, and the taller bars at the ends are the result of capping.

Q3. How does price per sq ft vary by property type? Apartments, Independent Houses and Villas have very similar price-per-sq-ft distributions. Their boxplots overlap almost completely, and the medians sit at about the same level (around ₹9,000 to ₹10,000 per sq ft). Property type does not appear to change the price per sq ft in this dataset.

Q4. Is there a relationship between property size and price? No clear relationship is visible. The scatter plot of size against price (sampled 5,000 points) forms a uniform cloud with no upward or downward trend. The correlation between size and price is close to zero. This is unusual for real estate data, where bigger homes normally cost more, and is discussed in Section 4.

Q5. Are there outliers in price per sq ft or property size?

Variable	Outliers (IQR method)	Share
Price per sq ft (raw data)	19,723	7.89%
Size (sq ft)	0	0%

All price-per-sq-ft outliers are on the high side, with values up to about ₹1,00,000 per sq ft. Size has no outliers. Outliers in price per sq ft were kept in the cleaned data, because they are a real part of the dataset and the Good Investment label is calculated from the median, which is not affected by them.

2.2 Location-based analysis

Q6. What is the average price per sq ft by state? Average price per sq ft is between about ₹12,900 and ₹13,200 in every state. Karnataka has the highest average and Rajasthan the lowest. The gap between the highest and lowest state is only about 2%, so state does not strongly change the price per sq ft in this data.

Q7. What is the average property price by city? Average price is also very similar across cities:

Group	Cities	Average price (lakh)
Top 15 (range)	Bangalore (highest), Surat, Kochi, Gaya, Mangalore and others	about 255 to 258
Bottom 5	Mumbai (252.1), Bilaspur (251.9), New Delhi (251.6), Trivandrum (251.3), Cuttack (250.8)	about 251 to 252

The total spread between the highest and lowest city is less than ₹7 lakh, which is less than 3% of the average price. Mumbai and New Delhi, which are among the most expensive cities in real life, appear at the bottom of this list. This is a clear sign that the dataset's city values do not follow real-market prices (see Section 4).

Q8. How is BHK distributed across cities? Each BHK type (1 to 5) makes up about 19% to 21% of properties in each of the top 10 cities. Overall there are about 50,000 properties per BHK type. No city has a dominant BHK type, which again differs from real markets, where cities usually have more 2 and 3 BHK homes than 5 BHK homes.

Q9. What is the median age of properties by locality?

- The dataset has 500 localities.
- Overall median property age is 18 years.
- Locality medians range from 16 years (Locality_443) to 21 years (Locality_72).
- The standard deviation of locality medians is only 0.78 years.

Locality has very little effect on property age. The youngest and oldest localities differ by only a few years.

Q10. What are the price trends in the top 5 most expensive localities? The top 5 localities by average price per sq ft are Locality_207, Locality_416, Locality_246, Locality_359 and Locality_387. Their median price per sq ft lies between about ₹9,800 and ₹10,400, and their median price between ₹257 and ₹267 lakh. The boxplots overlap almost completely.

The average-based ranking is pulled up by a small number of very high-priced properties (outliers up to ₹80,000 to ₹90,000 per sq ft). The medians are much closer together than

the averages.

Limitation: The brief asks for price trends over time, but the dataset has no sale date. It only has `Year_Built`, which is when the property was constructed, not when it was sold. A time trend cannot be shown, so this question is answered with a comparison of price distributions instead.

2.3 Feature relationships and correlation

Q11. How are numeric features correlated with each other? The correlation heatmap shows the following:

- **Age_of_Property and Year_Built** are almost perfectly negatively correlated (-1). This is expected, because age is calculated from the year built.
- **Price_in_Lakhs and Future_Price_5yr_Lakhs** are almost perfectly correlated (close to 1). This is expected, because the future price is calculated from the current price.
- **Price_per_SqFt and Price_in_Lakhs** have a moderate positive correlation, because both use the same price value.
- **Size_in_SqFt and Price_per_SqFt** have a negative correlation. Larger properties have a lower price per sq ft, which matches the Good Investment result below.
- **Good_Investment and Size_in_SqFt** have a positive correlation. Larger properties are more likely to be labelled Good Investment.
- Most other features (BHK, Floor, Nearby_Schools, Nearby_Hospitals, Amenities_Count) have correlations close to zero with each other and with price.

Q12. How do nearby schools relate to price per sq ft? The median price per sq ft for properties with 1 to 10 nearby schools lies between about ₹9,140 and ₹9,350. The correlation between the number of nearby schools and price per sq ft is -0.0001, which means there is no measurable relationship.

Q13. How do nearby hospitals relate to price per sq ft? The median price per sq ft for properties with 1 to 10 nearby hospitals lies between about ₹9,130 and ₹9,350. The correlation is also -0.0001. Like schools, nearby hospitals do not affect price per sq ft in this data.

Q14. How does price vary by furnished status? Furnished, Semi-furnished and Unfurnished properties have very similar price distributions. Their boxplots overlap closely, and medians are around ₹250 lakh for all three. Furnishing status does not appear to change the price.

Q15. How does price per sq ft vary by facing direction?

Facing	Median price per sq ft (₹)
West	9,308
East	9,233
South	9,224
North	9,219

The gap between the highest and lowest direction is less than ₹100 (about 1%). Facing direction has no meaningful effect on price per sq ft.

2.4 Investment, amenities and ownership analysis

Q16. How many properties belong to each owner type? The three owner types have almost equal counts, at about 83,000 properties each (roughly one-third of the dataset).

Note on the brief: The brief lists owner types as Individual, Builder and Agent. The dataset actually uses **Owner, Builder and Broker**. The answer above uses the dataset values.

Q17. How many properties are available under each availability status?

- Ready_to_Move: about 125,000 properties
- Under_Construction: about 125,000 properties

The two statuses are split almost exactly in half.

Note on the brief: The brief lists Available, Under Construction and Sold. The dataset has only **Ready_to_Move** and **Under_Construction**. No property is marked as Sold or Available.

Q18. Does parking space affect property price?

Parking	Count	Median price (lakh)	Median price per sq ft (₹)
No	125,456	253.0	9,246
Yes	124,544	254.0	9,245

The difference is about ₹1 lakh in median price, and almost nothing in price per sq ft. Parking does not affect price in this dataset.

Note on the brief: The brief describes Parking_Space as a number of parking spots. In the dataset it only contains Yes or No, so the effect of the number of spots cannot be tested.

Q19. How do amenities affect price per sq ft? Median price per sq ft for properties with 1 to 5 amenities lies between about ₹9,206 and ₹9,294. The correlation between amenities count and price per sq ft is 0.002, which is effectively zero. Amenities do not change price per sq ft in this data.

Q20. How does public transport accessibility relate to price per sq ft and investment potential?

Transport access	Median price per sq ft (₹)	Good Investment rate	Properties
Low	9,265	49.9%	83,287
Medium	9,269	49.8%	83,008
High	9,205	50.3%	83,705

Transport access has almost no effect on price per sq ft or on the share of Good Investment properties. The Good Investment rate is about 50% in every group, which is expected, because the label was created from a median split (see Section 1.4).

2.5 Summary of key findings

1. **Price and size are both almost uniformly distributed.** The dataset does not show the usual real-estate pattern where larger homes cost more.
2. **Location (state, city, locality) has very little effect** on price per sq ft or average price.
3. **Most property features have no measurable effect** on price per sq ft. These include schools, hospitals, facing direction, furnishing, parking, amenities and transport access.
4. **Size is the only feature with a clear relationship** to the Good Investment label, because larger properties have a lower price per sq ft.
5. **Locality ages are very similar** (median 16 to 21 years).

2.6 Data quality notes

These points should be stated in the report, because they affect how the findings should be read:

- The distributions of price, size, BHK, owner type and availability are almost perfectly flat. Real market data is usually skewed, so the dataset may be simulated or heavily processed.
- Several categories in the brief do not match the dataset (owner types, availability statuses, parking format).
- Crime rate and infrastructure score, which the brief asks to be analysed, are not in the dataset. Proxy features (Security, Public Transport, Nearby Schools, Nearby Hospitals) were used instead.

- Nearby_Schools and Nearby_Hospitals contain only the whole numbers 1 to 10, and Amenities_Count only 1 to 5.